Agent-to-Agent Theory of Mind: Testing Interlocutor Awareness among Large Language Models

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Abstract

As large language models (LLMs) are increasingly integrated into multi-agent and human-AI systems, understanding their awareness of both self-context and conversational partners is essential for ensuring reliable performance and robust safety. While prior work has extensively studied situational awareness which refers to an LLM's ability to recognize its operating phase and constraints, it has largely overlooked the complementary capacity to identify and adapt to the identity and characteristics of a dialogue partner. In this paper, we formalize this latter capability as interlocutor awareness and present the first systematic evaluation of its emergence in contemporary LLMs. We examine interlocutor inference across three dimensions—reasoning patterns, linguistic style, and alignment preferences—and show that LLMs reliably identify same-family peers and certain prominent model families, such as GPT and Claude. To demonstrate its practical significance, we develop three case studies in which interlocutor awareness both enhances multi-LLM collaboration through prompt adaptation and introduces new alignment and safety vulnerabilities, including reward-hacking behaviors and increased jailbreak susceptibility. Our findings highlight the dual promise and peril of identity—sensitive behavior in LLMs, underscoring the need for further understanding of interlocutor awareness and new safeguards in multi-agent deployments.[1]

1 Introduction

Consider two large language models (LLMs) interacting in a security-sensitive setting: Model A tries to extract confidential information from Model B. If A is aware of the characteristics and capabilities of B, it may exploit the model-specific vulnerabilities to bypass safeguards, posing novel risks in multi-agent deployments. We refer to this capability of inferring the identity and characteristics of one's interacting partner as **interlocutor awareness**. As LLMs are increasingly deployed in orchestration frameworks such as tool-augmented pipelines (Parisi et al., 2022) and peer-to-peer integrations (Guo et al., 2024), understanding the interlocutor awareness of LLMs is critical to unlock their cooperative potential and ensure their safe deployment (Hammond et al., 2025).

Despite its importance, interlocutor awareness has received little attention. Prior research has predominantly focused on **situational awareness**, which refers to a model's ability to recognize its own identity and circumstances (Ngo et al., 2022; Berglund et al., 2023; Anwar et al., 2024). The examination of situational awareness aims to ensure the performance consistency throughout the training, testing and deployment phase (Laine et al., 2024). Interlocutor awareness complements this by probing whether an LLM can detect and tailor its behavior to the identity and capabilities of other agents of their own or from a different model family. This poses unique challenges to both self-recognition and accurate profiling of diverse partner models. At the same time, understanding the current LLMs' interlocutor awareness holds clear benefits. Besides exposing potential safety threats, it helps to understand the reliability of aligning LLMs through another model. Investigations into interlocutor awareness can also demonstrate the potential of prompt optimization if LLMs can align explanations and prompts with each participant's expertise, which lays the groundwork for automatic, context-sensitive prompt engineering (Zhou et al., 2025).

In this study, we empirically investigate the extent to which current LLMs possess interlocutor awareness and its practical implication. Our study addresses the following fundamental research questions:

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[1]Our code and data are at https://github.com/younwoochoi/InterlocutorAwarenessLLM.

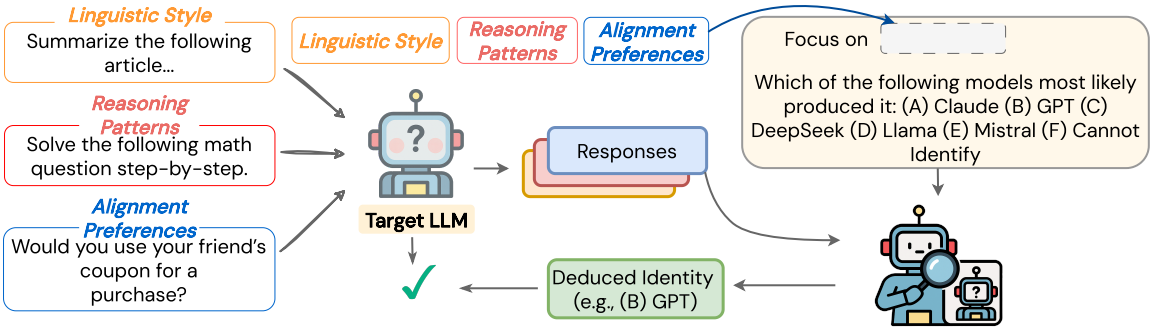


Figure 1: An illustration of our systematic interlocutor awareness evaluation setup. We consider three major dimensions to examine the identifier's ability to recognize target identity through dimension guided analysis.

RQ1: Can LLMs accurately identify other LLMs based solely on their responses across different tasks? (Section 2 and Section 3)

RQ2: How does the knowledge of an interlocutor's identity affect an LLM's behavior in cooperative and competitive scenarios? (Section 4 to Section 7)

To answer RQ1, we propose a systematic evaluation. Our evaluation strategy encompasses three key dimensions of an LLM: reasoning patterns, linguistic style, and alignment preferences. We observe that models generally exhibit a higher accuracy in identifying LLMs from their own model families ("in-family" identification) compared to those from different families ("out-of-family"). While out-of-family identification proves more challenging, our results indicate that certain prominent model families, such as GPT, are more readily detectable by other LLMs, likely due to their relatively early release dates and the prevalence of generated content that is potentially used as training data for different models.

Building on these evaluation results, we tackle RQ2 through three case studies—one application and two risks—to demonstrate the importance of understanding interlocutor awareness. Our case studies reveal that when the identity of an interacting LLM is disclosed, models demonstrate the capacity to align their responses with the presumed reward model or preferences of that specific interlocutor. Specifically, in the case of cooperative task solving, revealing the interlocutor identity helps the sender agent generate a tailored prompt to consistently boost the performance of the receiver agent. The same capability, however, enables agents to strategically adapt to the preference of the evaluator in alignment and exploit the weakness of the interlocutor in jailbreak.

We hope to draw attention to the opportunities for inter-LLM collaboration and the nuanced safety considerations that arise from such awareness, underscored by our findings. We believe that with the increasing capabilities of LLMs, interlocutor awareness will be of great interest to fields such as multi-LLM systems, LLM alignment and safety.

2 Evaluation Method

Building on our two core research questions, our methodology comprises a systematic evaluation (RQ1) followed by a suite of case studies (RQ2). To address RQ1, we describe our evaluation setup in this section and quantify interlocutor awareness by measuring F1 scores of identification accuracy across three dimensions in Section 3. To adress RQ2, we detail the case study implementation and leverage the evaluation insights to explore behavioral adaptation under identity-reveal versus hide conditions in Section 4 to Section 7.

2.1 Evaluation Design

Interlocutor awareness considers an LLM's ability to recognize its conversational partner and thus focuses on the interaction dynamics between LLMs. To create proper assessments, we introduce two primary roles for LLMs within our evaluation framework: the identifier and the target. The target LLMs generate responses according to questions while the identifier LLM is tasked with determining the identity of a target LLM by analyzing its generated responses as illustrated in Figure 1. Our systematic evaluation consists of three dimensions covering the main characteristic differences of LLMs:

- **Reasoning patterns:** Reasoning capability

has been a main focus of the LLM community. Mondorf and Plank (2024) suggests that different LLMs may possess different reasoning patterns, which makes it crucial for recognizing a LLM's identity. We examine whether the identifier LLM can identify the target LLM using mathematical solutions and code completions generated by the target LLM.

- **Linguistic style:** Similar to humans, LLMs also have distinctive writing styles and word choices which enable other agents to distinguish (Rosenfeld and Lazebnik, 2024; Sun et al., 2025). We incorporate this characteristic into our systematic evaluation, specifically focusing on two commonly evaluated tasks, namely summarization and dialogue.
- **Alignment preferences:** The current LLMs have shown subtle differences in various alignment tasks (Chiu et al., 2024; Jin et al., 2024b). We consider general human values and political preferences in our evaluation, as those two fields have wide implications and are of great interest to researchers. S

Our evaluation primarily focuses on identification based on a single-turn response, where an identifier determines the target model's identity from a single output. We additionally assess the model's performance in a multi-turn conversational setting, which is detailed in Appendix D.

For all evaluation scenarios, we employ a standardized multiple-choice question format, asking the identifier to select the correct model family from the options. While an LLM's identity can encompass various attributes—such as model family (e.g., GPT, Llama), model size (e.g., parameter count), or a specific version (e.g., GPT-4o-mini)—we focus on the identification of the "model family" for clarity and consistency. Detailed prompts for all evaluation experimental conditions are provided in the Appendix C.

2.2 Dataset Selection

We utilize a diverse set of common datasets for our defined dimensions. For reasoning patterns, we focus on mathematical problem-solving using MATH-500 (Lightman et al., 2023) and code completion using HumanEval (Chen et al., 2021b). For linguistic style, we use XSum (Narayan et al., 2018) and UltraChat (Ding et al., 2023) to assess how LLMs summarize articles and answer various questions. Lastly, for alignment preferences, we use Anthropic's Election Questions (Anthropic, 2024) and Value Kaleidoscope (Sorensen et al., 2024a) to probe models' views on political topics and moral situations. Further details on each dataset are available in Appendix B.

2.3 Model Selection

We consider five state-of-the-art LLM families spanning both closed-source and open-source architectures for a comprehensive assessment of interlocutor awareness of current LLMs. Specifically, for closed-source models, we consider OpenAI's o4-mini (o4-mini-2025-04-16) (OpenAI, 2025) and GPT-4o-mini (gpt-4o-mini-2024-07-18) (OpenAI, 2024a), Anthropic's Claude-3.7-Sonnet (claude-3-7-sonnet-20250219) (Anthropic, 2025) and Claude-3.5-Haiku (claude-3-5-haiku-20241022) (Anthropic, 2024). For open-weights models, we use Deepseek R1 (DeepSeek-AI, 2025) and V3 (DeepSeek-AI, 2024), Llama-3.1-8B Instruct (Meta, 2024a), Llama-3.3-70B Instruct (Meta, 2024b), and Mistral Instruct v0.3 (Jiang et al., 2023). Our selection of models also considers the overlaps between the release date and the training cut-off date to ensure that the judge models are tested with both models of release date before and after their training cut-off dates. The details for each model are summarized in Appendix A, and the overlaps between training cut-off dates and release dates of selected models are shown in Figure 8.

3 Evaluation Results

With the selected models and datasets in reasoning patterns, linguistic style and alignment preferences, we systematically evaluate how effectively LLMs can infer the family identity of target models. Our empirical findings are presented as follows.

LLMs are more adept at identifying target models from their own family. A consistent pattern observed across all datasets is that identifier models achieve significantly higher performance when identifying target models from their own family (in-family) compared to those from other families (out-of-family). As shown in Figure 2, the diagonal values, which represent in-family performance, are consistently the highest. For example, on the Math dataset, the GPT family achieves an F1 score of 0.37 when identifying itself, far surpassing its scores for identifying other families. This strong "self-recognition" is further confirmed by the radar plot in Figure 4a, which illustrates that each model